

QA LAB BRASIL

Reference guide - QA Lab Brasil

Docker & Containers for QA

Images, containers, Dockerfiles, volumes, ports and why containers help reproduce test environments.

Image vs container

An image is an immutable package with an application and its dependencies. A container is a running instance of that image, isolated from the rest of the system.

Dockerfile and execution

A Dockerfile describes how to build an image. Ports, volumes, environment variables and networks control communication and data persistence.

Use in QA

Containers help create reproducible dependencies, temporary databases, mocks, services and local environments close to the pipeline.

Considerations

Different versions, persisted data, networking, resources and secrets can produce different behavior. Record the image/tag used in evidence when relevant.

QA Tip: If a bug occurs only in one environment, comparing container image, configuration and variables can reveal important differences.

Official: <https://www.docker.com/>